

Name _____

Quiz 6

Recall the problem from the Prairie Learn; its complete statement and solution is provided on the other side of this sheet. For this quiz, you need to fill in the solution for a similar problem.

Problem statement. Momo is planning another road trip from mile marker 1 to miles marker n . However, she can now go three miles without stopping! Otherwise, the problem is the same: she is currently at mile marker 1, and she pays $HotelCosts[i]$ wherever she stops for the night. Determine the minimum possible total cost of all hotels that Momo stays in during her trip.

1. Suppose $HotelCosts = [0, 1, 4, 3, 2, 0]$. At which mile marker(s) does an optimal trip stop?

2. What is the cost of the optimal trip from (a)? _____

3. Complete the English recursive subproblem definition for this problem.
Let $MinCost(i)$ denote...

4. For what value of i does $MinCost(i)$ give the solution to the full problem? _____

5. Fill in the recurrence for this problem. You may not need every row given.

$$MinCost(i) = \left\{ \begin{array}{ll} \text{_____} & \text{if } \text{_____} \\ \text{_____} & \text{if } \text{_____} \\ \text{_____} & \text{if } \text{_____} \\ \text{_____} & \text{if } \text{_____} \\ \text{_____} & \text{if } \text{_____} \end{array} \right.$$

6. Complete the pseudocode for a memoized algorithm to solve the problem.

```
ComputeMinTotalCost( $HotelCosts[1..n]$ )  
  let array  $MinCost[1..n]$ 
```

Problem from Prairie Learn

Momo is planning a road trip along a highway with n mile markers, numbered from 1 through n . Momo is currently at mile marker 1, and she would like to travel to mile marker n . However, Momo has a very low attention span and gets exhausted easily, so she can only drive 1 or 2 miles forward on a single day.

Luckily, there is a hotel right next to each mile marker; the costs of these hotels are provided in the array $HotelCosts[1..n]$. If Momo stops at mile i at the end of the day, she will stay overnight in a hotel that costs $HotelCosts[i]$. Determine the minimum possible total cost of all the hotels that Momo stays in during her road trip.

Momo does not have to stay in a hotel at mile 1 or mile n , so $HotelCosts[1] = HotelCosts[n] = 0$. She cannot stay in the same hotel two nights in a row. For example, if $HotelCosts = [0, 1, 374, 3, 0]$, an optimal trip stops at miles 2 and 4, for a total cost of $1 + 3 = 4$.

- **English recursive subproblem definition.** Let $MinCost(i)$ denote the minimum possible cost of traveling from mile 1 to mile i , including the cost of the hotel at mile i .
- **Recurrence.** The trip to mile i must come from mile $i - 1$ or mile $i - 2$. Thus,

$$MinCost(i) = \begin{cases} 0 & \text{if } i = 1, \\ HotelCosts[2] & \text{if } i = 2, \\ HotelCosts[i] + \min\{MinCost(i - 1), MinCost(i - 2)\} & \text{if } 3 \leq i \leq n. \end{cases}$$

- **Memoized algorithm.**

```
ComputeMinTotalCost(array  $HotelCosts[1..n]$ ):  
  let array  $MinCost[1..n]$   
   $MinCost[1] = 0$   
   $MinCost[2] = HotelCosts[2]$   
  for  $i = 3$  to  $n$ :  
     $MinCost[i] = HotelCosts[i] + \min\{MinCost[i - 1], MinCost[i - 2]\}$   
  
  return  $MinCost[n]$ 
```